

Combined ratio estimation example

Using IPUMS ACS 2024 Data for New York State

Introduction

In class, we studied the **ratio estimator** under stratified sampling (STRS). In study, I will apply that theory to real survey data from the American Community Survey (ACS), accessed through [IPUMS USA](#).

In this activity we use **annual hours worked** as our auxiliary variable and **wage income** as our variable of interest. We treat the full filtered ACS dataset as the population, draw an stratified sample from it, and compare our class theory to the results from the survey package.

Setup

```
library(tidyverse)
library(ipumsr)
library(survey)
```

```
ddi <- read_ipums_ddi("usa_00011.xml")
data <- read_ipums_micro(ddi)
```

Prepare the Population

We filter the ACS data to private wage and salary workers in New York State with positive hours and wage income, removing the top and bottom 5% of earners to reduce the influence of extreme outliers. We then construct **annual hours worked** as our auxiliary variable. For this question we will consider this the population.

```
ny_pop <- data |>
  filter(
    STATEFIP == 36,      # New York
    UHRSWORK > 0,
    INCWAGE > 0,
    INCWAGE < 999998,    # 999998/999999 are IPUMS N/A codes
    WKSWORK1 > 0,
    CLASSWKR == 2        # private wage/salary workers
  ) |>
  filter(
    INCWAGE < quantile(INCWAGE, 0.95),
    INCWAGE > quantile(INCWAGE, 0.05)
  ) |>
  mutate(annual_hours = UHRSWORK * WKSWORK1)

nrow(ny_pop)
```

```
[1] 86436
```

Take a stratified sample

```
# Add stratum population sizes for FPC
stratum_sizes <- ny_pop |>
  count(STRATA) |>
  rename(N_h = n)

ny_strat_sample <- ny_pop |>
  left_join(stratum_sizes, by = "STRATA") |>
  group_by(STRATA) |>
  sample_frac(0.01) |>
  ungroup()

# Set up stratified design using existing STRATA variable
```

```
svy_strat <- svydesign(
  ids      = ~1,
  strata   = ~STRATA,
  fpc      = ~N_h,
  data     = ny_strat_sample
)
```

Combined ratio estimator

Using R package

```
## Using R package
t_x <- sum(ny_pop$annual_hours)

ratio_combined <- svyratio(
  numerator   = ~INCWAGE,
  denominator = ~annual_hours,
  design      = svy_strat
)

print(ratio_combined)
```

```
Ratio estimator: svyratio.survey.design2(numerator = ~INCWAGE, denominator = ~annual_hours,
  design = svy_strat)
```

Ratios=

	annual_hours
INCWAGE	31.33646

SEs=

	annual_hours
INCWAGE	0.6156506

```
predict(ratio_combined, total = t_x)
```

\$total

	annual_hours
INCWAGE	4916972852

\$se

	annual_hours
INCWAGE	96601118

Manually

```
# Compute stratum-level statistics
stratum_stats <- ny_strat_sample |>
  group_by(STRATA) |>
  summarise(
    n_h      = n(),
    N_h      = first(N_h),
    ybar_h   = mean(INCWAGE),
    xbar_h    = mean(annual_hours),
    s_y2_h   = var(INCWAGE),
    s_x2_h   = var(annual_hours),
    s_xy_h   = cov(INCWAGE, annual_hours)
  )

# Global B_hat using stratified estimators
t_hat_y_str <- sum(stratum_stats$N_h * stratum_stats$ybar_h)
t_hat_x_str <- sum(stratum_stats$N_h * stratum_stats$xbar_h)
B_hat      <- t_hat_y_str / t_hat_x_str

# Stratum residual variances using global B_hat
stratum_stats <- stratum_stats |>
  mutate(
    s_eh2 = s_y2_h - 2*B_hat*s_xy_h + B_hat^2*s_x2_h
  )

# Manual variance using our formula
V_manual <- (t_x/t_hat_x_str)^2 * sum(
  stratum_stats$N_h^2 * (1 - stratum_stats$n_h / stratum_stats$N_h) *
  stratum_stats$s_eh2 / stratum_stats$n_h
)

t_hat_ycr <- B_hat * t_x
```

Comparison:

```
cat("=== Point estimates ===\n")
```

```
=== Point estimates ===
```

```
cat("Manual t_hat_yc:  ", t_hat_yc, "\n")
```

Manual t_hat_yc: 4916972852

```
cat("Survey t_hat_yc:  ", as.numeric(predict(ratio_combined, total = t_x)[1]), "\n\n")
```

Survey t_hat_yc: 4916972852

```
cat("=== Standard errors ===\n")
```

=== Standard errors ===

```
cat("Manual SE:  ", sqrt(V_manual), "\n")
```

Manual SE: 96601118

```
cat("Survey SE:  ", as.numeric(predict(ratio_combined, total = t_x)[2]), "\n")
```

Survey SE: 96601118